

Chapter revision

• The circle

Every rule of the chapter on one page, and the strategy for choosing between them.

LESSON 60

1 × 40 MIN

GEOMETRY 8, CHAPTER REVISION

STAGE 9 · 7.1–7.3

LESSON CLOCK

5'

15'

15'

5'

Name every rule from memory	5 min
The summary table	15 min
Mixed problems	15 min
What to check tomorrow	5 min

LEARNING OBJECTIVES

- Recall every circle rule from Quarter IV.
- Choose the right rule from the position of the vertex.
- Combine the chord property with Pythagoras.
- Work multi-step circle problems.

TEXTBOOK

National	Темы 55–60, pp. 128–148
Cambridge	Learner’s Book pp. 142–162
Workbook	Workbook 7.1–7.3

01

Explanation

Everything on one page

COMPARISON	THE LINE IS	COMMON POINTS
$d > r$	outside — it misses	0
$d = r$	a tangent	1
$d < r$	a secant	2

THE RULES, IN ORDER OF USE

- Tangent $\perp$ radius** at the point of tangency — every tangent problem has a right triangle.
- Two tangents from a point are equal**, and the line to the centre bisects the angle.
- The perpendicular from the centre bisects a chord**: $r^2 = d^2 + (a/2)^2$.
- Central angle = arc**; **inscribed angle = $\frac{1}{2}$ arc**.
- Angle in a semicircle = 90°** ; angles on the same arc are equal.

6. **Cyclic quadrilateral**: opposite angles add to 180° .

7. Vertex **inside**: half the sum of the arcs. Vertex **outside**: half the difference.

8. **Tangent–chord angle** = half the intercepted arc.

9. $C = 2\pi r$, $S = \pi r^2$, $\text{arc} = \frac{\alpha}{360^\circ} \cdot 2\pi r$, $\text{sector} = \frac{\alpha}{360^\circ} \cdot \pi r^2$.

How to choose

Every angle question in this chapter is answered by asking one thing first:

Where is the vertex?

At the centre → the arc. On the circle → half the arc. Inside → half the sum. Outside → half the difference. That question alone chooses the rule in four cases out of five.

Every length question is answered by a different single question: **where is the right triangle?** It is at the point of tangency, or under the perpendicular from the centre to a chord. Draw it in, and Pythagoras finishes the job.

TWO CHECKS BEFORE YOU WRITE THE ANSWER

An angle in a circle problem is almost never more than 180° , and a chord is never longer than the diameter. Both catch a great many slips.

02 Worked examples

EXAMPLE 1 A chord of 12 cm is drawn in a circle of radius 10 cm. Find its distance from the centre and the central angle it subtends.

① $r^2 = d^2 + 6^2$
The perpendicular bisects the chord.

② $d = \sqrt{100 - 36} = 8 \text{ cm}$

③ $\sin(\alpha/2) = \frac{6}{10} = 0.6$
In the right triangle.

④ $\alpha/2 \approx 36.9^\circ, \alpha \approx 73.7^\circ$

ANSWER

$d = 8 \text{ cm}, \alpha \approx 74^\circ$

EXAMPLE 2 From an external point P , a tangent of length 8 touches a circle of radius 6. Find OP and $\angle OPT$.

① $OP^2 = 64 + 36 = 100$
Tangent $\perp$ radius.

② $OP = 10$

③ $\sin \angle OPT = \frac{6}{10} = 0.6$

④ $\angle OPT \approx 36.9^\circ$

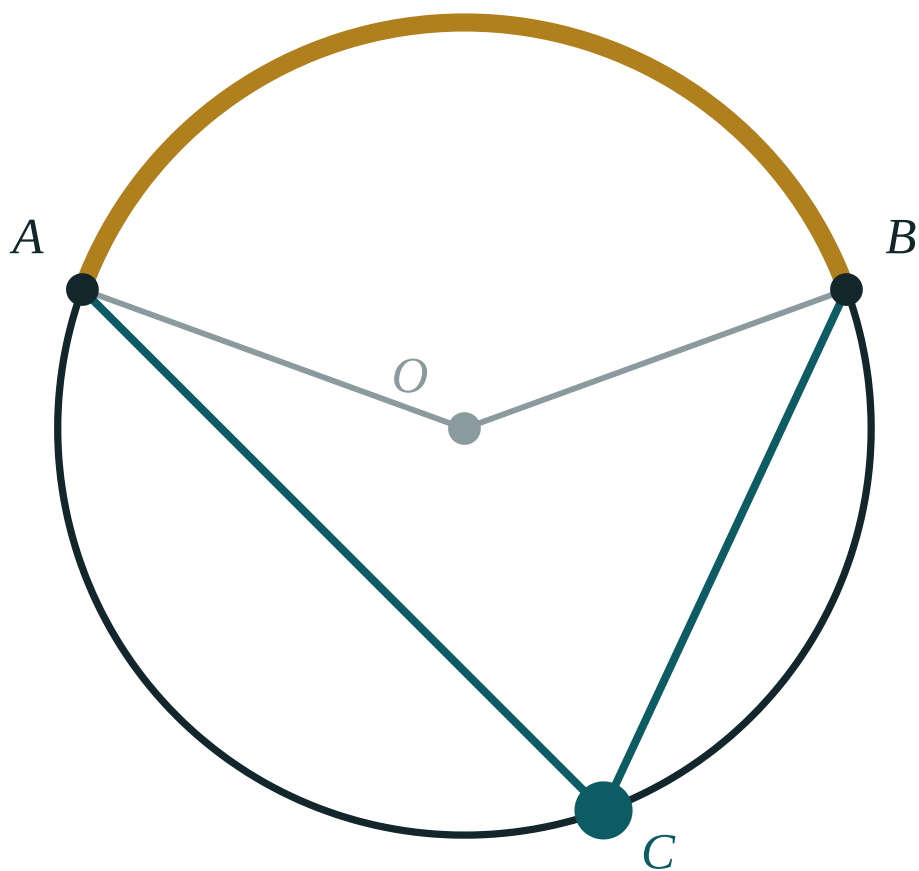
ANSWER

$OP = 10, \angle OPT \approx 37^\circ$

03 Interactive model

Use the model to re-check each rule of the chapter before tomorrow's paper.

All the circle angles



central $\angle AOB$ 140°

inscribed $\angle ACB$ 70°

ratio 2

arc AB 140°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Tangent	Urinma	Касательная
Secant	Kesuvchi	Секущая
Chord	Vatar	Хорда
Central angle	Markaziy burchak	Центральный угол
Inscribed angle	Ichki chizilgan burchak	Вписанный угол
Cyclic quadrilateral	Ichki chizilgan to'rtburchak	Вписанный четырёхугольник
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Revision	Takrorlash	Повторение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A tangent meets the radius at:

An inscribed angle on a 140° arc is:

$r = 13$, chord 10 . Its distance from the centre is:

Vertex outside the circle means:

half the sum

half the difference

the arc itself

half the arc

In a cyclic quadrilateral $\angle A + \angle C$ equals:

90°

180°

360°

it varies

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 10, d = 6$. Find the chord.

ANSWER 16

2. $OP = 13, r = 5$. Find the tangent length.

ANSWER 12

3. A central angle is 100° . Find the inscribed angle on the same arc.

ANSWER 50°

4. AB is a diameter, C on the circle. Find $\angle ACB$.

ANSWER 90°

5. In a cyclic quadrilateral $\angle B = 72^\circ$. Find $\angle D$.

ANSWER 108°

6. Find the area of a circle of radius 7 (in terms of π).

ANSWER 49π

7. Two chords meet inside, arcs 50° and 70° . Find the angle.

ANSWER 60°

Medium · the standard question

7 PROBLEMS

1. $r = 17$, chord 16. Find d .

ANSWER 15

2. A tangent from P is 20 and $r = 15$. Find OP .

ANSWER 25

3. Two secants meet outside, arcs 120° and 40° . Find the angle.

ANSWER 40°

4. A tangent–chord angle cuts off a 96° arc. Find it.

ANSWER 48°

5. Find the arc length of a 135° arc in a circle of radius 8.

ANSWER 6π

6. AB is a diameter and $\angle ABC = 25^\circ$. Find $\angle BAC$ and arc AC .

ANSWER 65° , arc 50°

7. $r = 25$, chord 30. Find d and the sector area of the corresponding minor sector.

ANSWER $d = 20$; $\sin(\alpha/2) = 0.6$, $\alpha \approx 73.7^\circ$, sector ≈ 402

Hard · stretch and reason

7 PROBLEMS

1. Two parallel chords of a circle of radius 25 are 14 and 40 long. Find the two possible distances between them.

ANSWER $24 + 15 = 39$ or $24 - 15 = 9$

2. Two tangents from P touch a circle of radius 7 with $\angle APB = 60^\circ$. Find OP , PA and the area of the kite $OAPB$.

ANSWER $OP = 14$, $PA = 7\sqrt{3}$, area $49\sqrt{3} \approx 84.9$

3. A chord subtends a central angle of 120° in a circle of radius 10. Find its length, its distance from the centre, and the area of the minor segment.

ANSWER chord $10\sqrt{3}$, $d = 5$, segment $\frac{100\pi}{3} - 25\sqrt{3} \approx 61.4$

4. AB is a diameter of a circle of radius 13, and C is on the circle with $AC = 10$. Find BC and the area of triangle ABC .

ANSWER $BC = 24$, area 120

5. A circle passes through $(0;0)$, $(6;0)$ and $(0;8)$. Find its centre and radius.

ANSWER centre $(3; 4)$, $r = 5$ — the angle at the origin is right, so the hypotenuse is a diameter

6. Two circles of radius 10 intersect, with their centres 16 apart. Find the length of the common chord.

ANSWER $12 - \sqrt{100 - 64} = 6$, doubled

7. Explain in one sentence why every tangent problem contains a right-angled triangle.

ANSWER The radius to the point of tangency is perpendicular to the tangent, so that radius, the tangent segment and the line to the centre always form a right triangle.

07 Homework

Homework — 5 problems

Revise for tomorrow's control work. Learn the nine rules; work the mixed problems.

1. Write out the nine rules of the chapter from memory, then check them against the list.

2. $r = 20$, chord 24. Find d .
3. A tangent from P is 24 and $r = 10$. Find OP .
4. Two chords meet inside, arcs 88° and 52° . Find the angle.
5. AB is a diameter of a circle of radius 15 and $AC = 18$. Find BC .